

Cybersecurity

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# Evidence Acquisition and Reports

Lesson 21.4



# Learning Objectives

By the end of today's lesson, you will be able to:

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- 1 Use Autopsy to view and gather evidence from Peter's emails.
- 2 Use data exports to analyze email messages offline
- 3 Use data exports to analyze SMS messages offline.
- 4 Use Autopsy to extract GPS data and identify Wi-Fi locations.
- 5 Prepare a preliminary report using evidence worksheets.



Welcome



Let's Recap



# Recap

In the last lesson, we covered:

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01

Identifying methods used in smartphone forensics.

02

Examining databases and file structures of an iPhone directory image.

03

Locating and documenting evidence of ownership for an iPhone image.

04

Using Autopsy to view and tag evidence of an iPhone image.

05

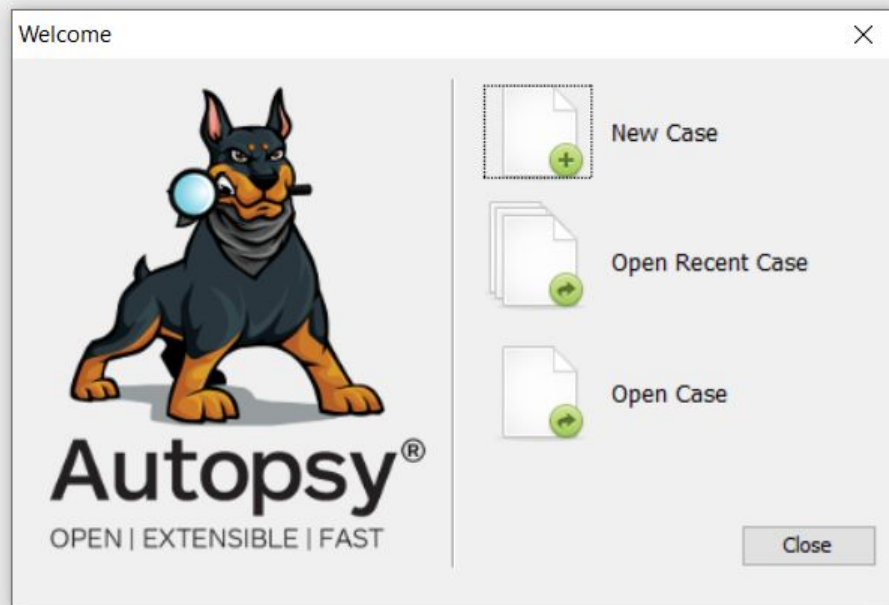
Extracting data for use in other applications (logs, text, pictures, video, audio).

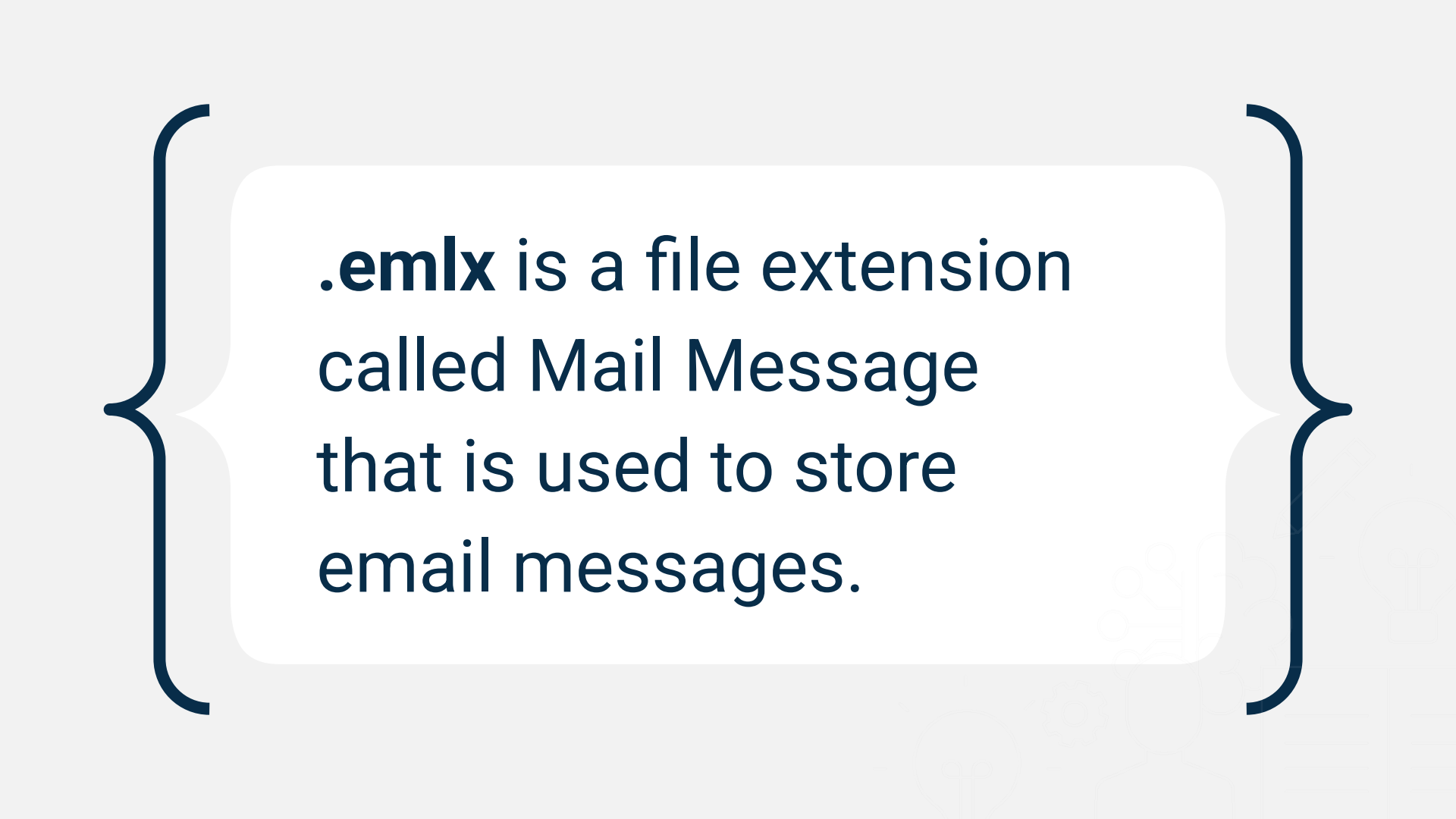


# Overview

# Overview

As a digital forensics investigator, it's critical that you understand how to use tools such as **Autopsy** and how to export data so that other team members can perform offline analysis of evidence.

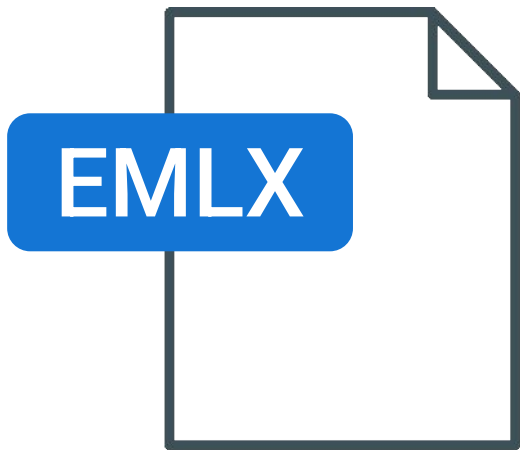




**.emlx** is a file extension  
called Mail Message  
that is used to store  
email messages.

# Overview

EMLX files are often referred to as Apple Mail files because they are created with Apple's mail program to store plain text files for a single message.





# Activity

## Email export

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In this activity, you will continue your role as a digital forensics investigator.

- Your task is to export the email database for offline analysis using examination tools outside of Autopsy.

**Suggested Time**  
15 minutes

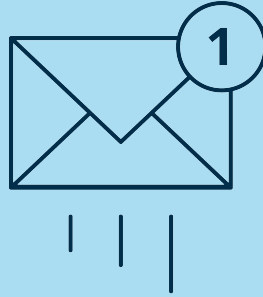


Time's Up!  
Let's review



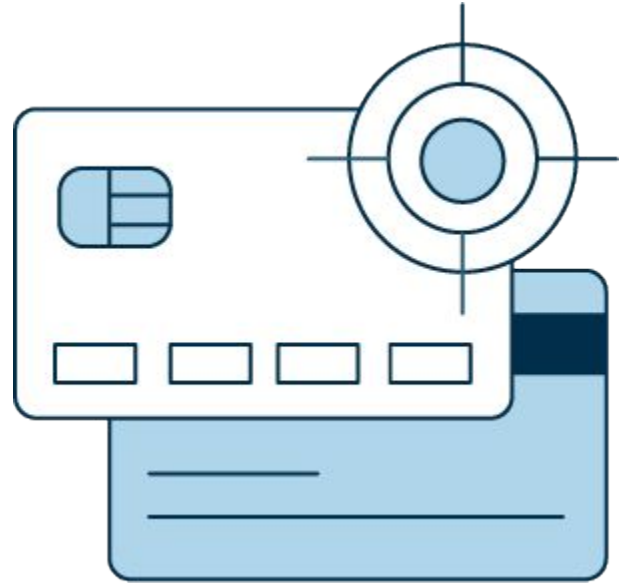
Questions?





# Peter's **Email Evidence**

Now we'll use offline analysis to uncover details about the plot of the crime and possibly identify the co-conspirators and their roles.



# Email Evidence

We can extract the following evidence from emails:

01

Sender's email address

02

Sender's IP address

03

Internet service provider (ISP)

04

User client (email app)

05

Location information



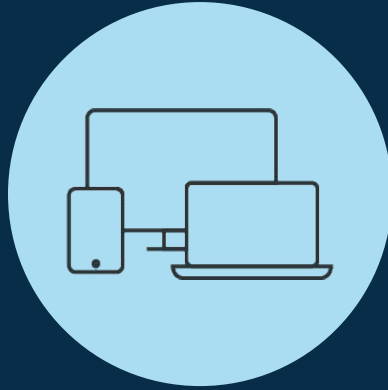
# Examining Emails in Kali Linux

Displaying email messages in the Mail/MessageData folder is as simple as running the `ls -l` command. From there, we'll use `nano` to display the contents of each EMLX file.



```
root@kali:~/casedata/Pure_Gold_5/Export/9-Mail/MessageData# ls -ltr
total 88
drwxr-xr-x 2 root root 4096 Jan 5 11:53 1
drwxr-xr-x 2 root root 4096 Jan 5 11:53 10
drwxr-xr-x 2 root root 4096 Jan 5 11:53 11
drwxr-xr-x 2 root root 4096 Jan 5 11:53 12
drwxr-xr-x 2 root root 4096 Jan 5 11:53 13
drwxr-xr-x 2 root root 4096 Jan 5 11:53 14
drwxr-xr-x 2 root root 4096 Jan 5 11:53 15
drwxr-xr-x 2 root root 4096 Jan 5 11:53 16
drwxr-xr-x 2 root root 4096 Jan 5 11:53 17
drwxr-xr-x 2 root root 4096 Jan 5 11:53 18
drwxr-xr-x 2 root root 4096 Jan 5 11:53 19
drwxr-xr-x 2 root root 4096 Jan 5 11:53 2
drwxr-xr-x 2 root root 4096 Jan 5 11:53 20
drwxr-xr-x 2 root root 4096 Jan 5 11:53 21
drwxr-xr-x 2 root root 4096 Jan 5 11:53 22
drwxr-xr-x 2 root root 4096 Jan 5 11:53 3
drwxr-xr-x 2 root root 4096 Jan 5 11:53 4
drwxr-xr-x 2 root root 4096 Jan 5 11:53 5
drwxr-xr-x 2 root root 4096 Jan 5 11:53 6
drwxr-xr-x 2 root root 4096 Jan 5 11:53 7
```

**Search within these  
directories for eml  
files**

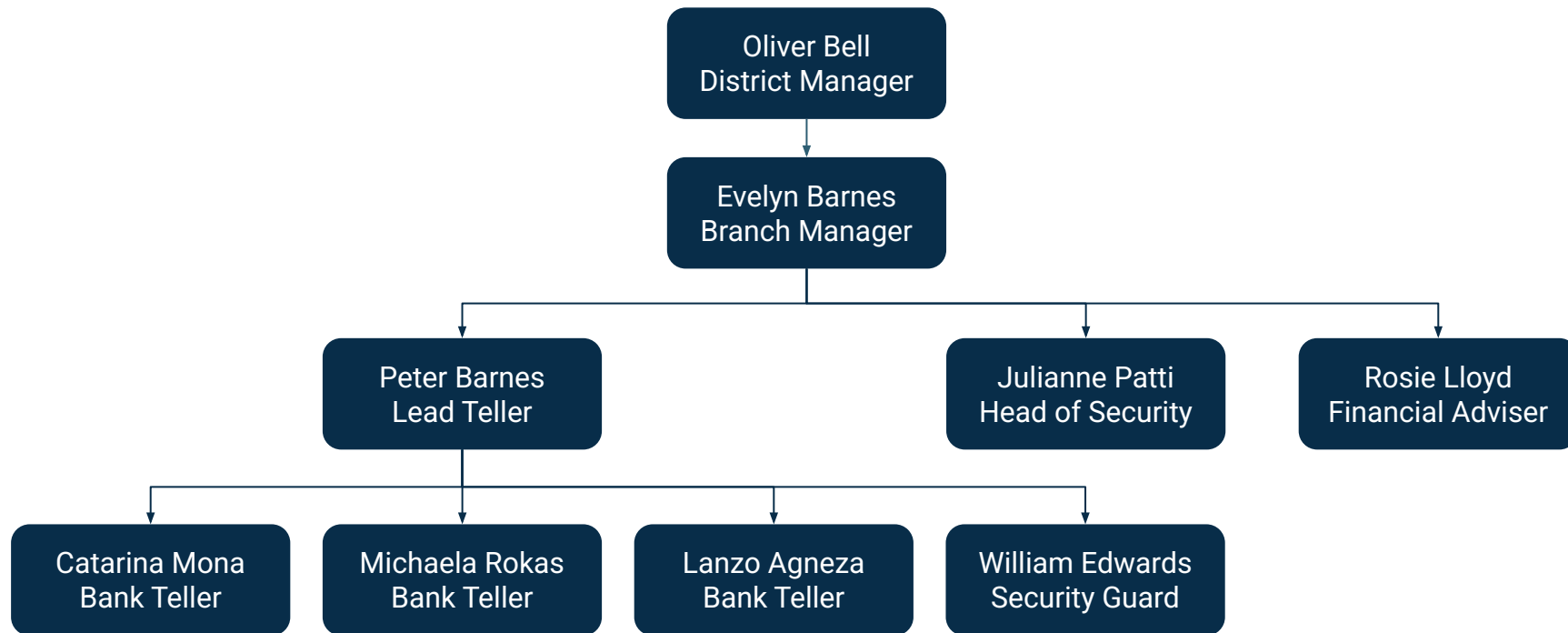


# Instructor Demonstration

Examine emails in Kali Linux

# Evidence in the Emails

This diagram provides a high-level overview of the threat actors' involvement in the Pure Gold Credit Union case. As you analyze the emails, use this as a road map to tie Peter's associates to each incident.



# Correspondence Evidence Worksheet

In the next activity, you will use the **Correspondence Evidence Worksheet** to document evidence from the communications found on Peter's phone (i.e., phone calls, emails, text messages).

**You will fill out the following information:**

|                                           |                                                                                                            |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Artifact numbers to help organize records | Number each record as you enter it into the worksheet.                                                     |
| Timestamp                                 | The time the email was sent or received.                                                                   |
| Header information                        | These should include names of the individuals involved, their email addresses, and the email subject line. |
| Key information                           | A summary of the email contents.                                                                           |
| Evidence location                         | The data source (e.g., email, SMS, phone call, etc.).                                                      |



# Activity

## Peter's email evidence

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In this activity, you will continue your role as a digital forensics investigator and dig deeper into the iPhone image to find more information about Peter's involvement in the crime.

**Suggested Time**  
45 minutes





Time's Up!  
Let's review



Questions?





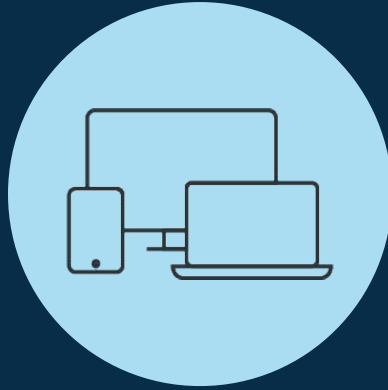
Break  
15 mins



# Peter's SMS Messages

We'll continue to search for more evidence by examining Peter's **SMS messages**.





# Instructor Demonstration

## Forensic analysis of SMS messages



# Activity

## Peter's SMS messages

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In this activity, you will work with your group to examine Peter's SMS messages and gather more information about the case.

**Suggested Time**  
15 minutes





Time's Up!  
Let's review



Questions?





# Activity

## The final report

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In this activity, you will start preparing a report of your group's findings. You will continue this report as part of your challenge.

**Suggested Time**  
25 minutes





Time's Up!  
Let's review



Questions?





Let's Recap

# Module Recap

We've examined a lot of information and done the following:

01

Performed mobile forensics analysis and compiled details of Peter's iPhone.

02

Searched through numerous files and directories on the iPhone image.

03

Tagged and categorized evidence relevant to the case.

04

Created custom tags.

# Module Recap

We've examined a lot of information and done the following:

05

Extracted data for offline analysis using the export function.

06

Examined and documented Peter's emails.

07

Examined and documented Peter's SMS messages.

# Lesson Wrap-Up

Key takeaways from this lesson

01

Digital forensics is a field dedicated to identifying, extracting, preserving, and reporting against information obtained from computer and network systems.

02

Digital forensics relies on the expertise of examiners to analyze and interpret data retrieved by trusted digital forensics examination tools.

03

Investigative teams may be spread across several time zones, so it's important to follow a baseline time zone as indicated by case file.



You'll continue your report and work on another **bonus assignment** for this week's challenge.



Questions?





# Next Class

Introduction to security within the organization



Thank You